Audio Visualizer: Sizing the LED Loudness Ladder

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Ten LEDs per channel that light in sequence with loudness. No driver IC, just transistors, resistors, and half an LM358 per channel. This is the math behind the resistor values.

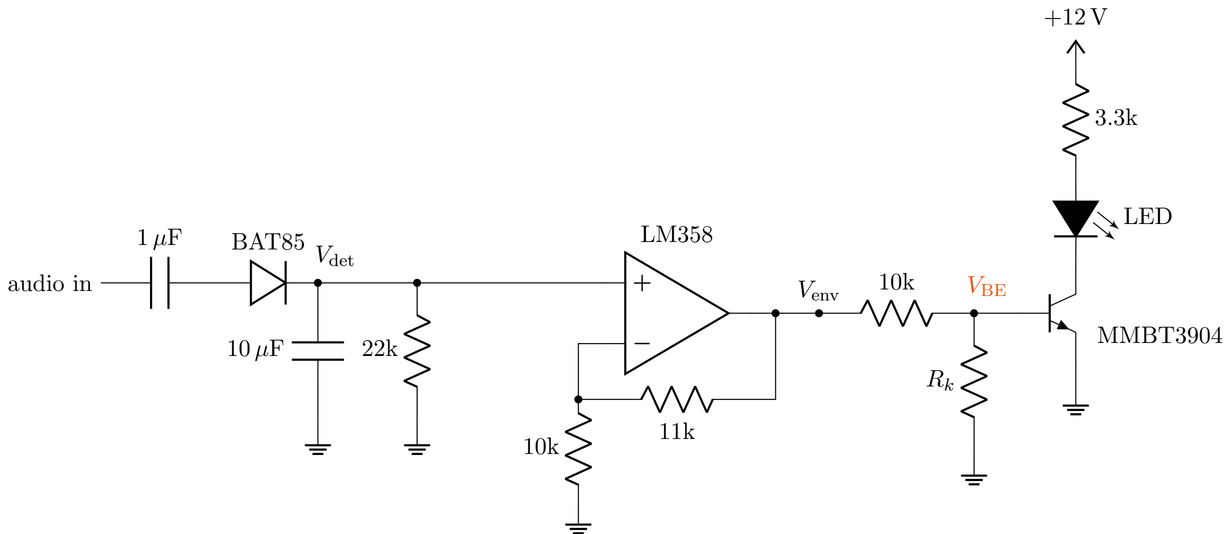

Figure 1: Front end and one of ten ladder stages (one channel of two). Only R_k changes from stage to stage. LED k turns on when its base hits $V_{\rm BE}$; the threshold V_k is not a node, just the $V_{\rm env}$ value that gets the base there.

Getting a loudness voltage

Audio is AC; the ladder needs a DC voltage that tracks volume. A $1\,\mu$F cap couples the signal in, a Schottky rectifies it, a $10\,\mu$F cap holds the peak, and a $22\,\mathrm{k}\Omega$ bleed lets it fall ($\tau = 22\,\mathrm{k}\Omega \times 10\,\mu\mathrm{F} \approx 0.22\,\mathrm{s}$). That's $V_{\rm det}$.

I call $V_{\rm det} = 5\,\mathrm{V}$ clipping. That's an educated guess: the speakers can theoretically swing 6 V, but the Schottky eats $\sim$0.3 V right after the coupling cap and the other parts in series take their share. This display is purely cosmetic, so being off by a little here costs nothing. 5 V it was.

Log spacing, because hearing is logarithmic

Equal steps in perceived loudness are equal voltage *ratios*, not equal volts. So the thresholds are spaced in dB, the same call the LM3915 bar driver this circuit replaces makes. 30 dB of range, 13 steps, 2.5 dB each:

$$L_k = -30 + 2.5\,(k-1)\ \mathrm{dB}, \qquad V_k = V_{\mathrm{fs}} \cdot 10^{L_k/20}. \tag{1}$$

Filling the 12 V rail

The detector tops out at 5 V but the board runs on 12. An LM358 on a 12 V rail can only swing to about 10.5 V, so I picked the gain to land full scale exactly there: $A = 10.5/5 = 2.1$, built as $1 + 11\mathrm{k}/10\mathrm{k}$. Full scale and clipping are the same voltage on purpose, which makes the top LED the clip light. The op amp also buffers the detector: the ladder pulls its current from the op amp instead of draining the peak-hold cap.

One transistor, three resistors per LED

Each stage is an NPN with a grounded emitter and an LED with its 3.3 kΩ current limiter on the collector. Two more resistors form the base divider that decides when the stage turns on:

$$V_k = V_{\mathrm{BE}}\left(1 + \frac{R_{\mathrm{top}}}{R_k}\right) \quad\Longrightarrow\quad R_k = R_{\mathrm{top}}\,\frac{V_{\mathrm{BE}}}{V_k - V_{\mathrm{BE}}}. \tag{2}$$

A divider can only divide, so nothing below $V_{\mathrm{BE}} \approx 0.65$ V is reachable. Any division under that line simply can't be built. The 13 came from trial and error: with 13 divisions, exactly three (0.33, 0.44, 0.59 V) fall under V_{BE}, and the 10 LEDs I wanted survive. Usable range: −22.5 to 0 dB.

Every R_{top} is 10 kΩ. That leaves exactly one resistor to calculate per LED, and it caps each divider at about a milliamp at full scale. The ideal R_k from Eq. (2) rounds to the nearest standard E24 value, which shifts the threshold slightly; the error column is how far the resulting threshold lands from the ideal V_k. Worst case is 3.8%, and adjacent thresholds are 33% apart, so the turn-on order never flips:

LED	L_k	R_k ideal	R_k (E24)	threshold	error
1	−22.5 dB	47.31 kΩ	47 kΩ	0.79 V	+0.1%
2	−20.0 dB	16.25 kΩ	16 kΩ	1.06 V	+0.6%
3	−17.5 dB	8.66 kΩ	9.1 kΩ	1.36 V	−2.6%
4	−15.0 dB	5.34 kΩ	5.1 kΩ	1.93 V	+3.1%
5	−12.5 dB	3.53 kΩ	3.6 kΩ	2.46 V	−1.4%
6	−10.0 dB	2.43 kΩ	2.4 kΩ	3.36 V	+1.1%
7	−7.5 dB	1.72 kΩ	1.8 kΩ	4.26 V	−3.8%
8	−5.0 dB	1.24 kΩ	1.2 kΩ	6.07 V	+2.8%
9	−2.5 dB	900 Ω	910 Ω	7.79 V	−1.0%
10	0.0 dB	660 Ω	680 Ω	10.21 V	−2.8%

Collector resistors: $R_C = (12 - 2.0 - 0.2)\ \mathrm{V}/3\ \mathrm{mA} \approx 3.3\ \mathrm{k\Omega}$ per LED. The board carries two identical channels, left and right.

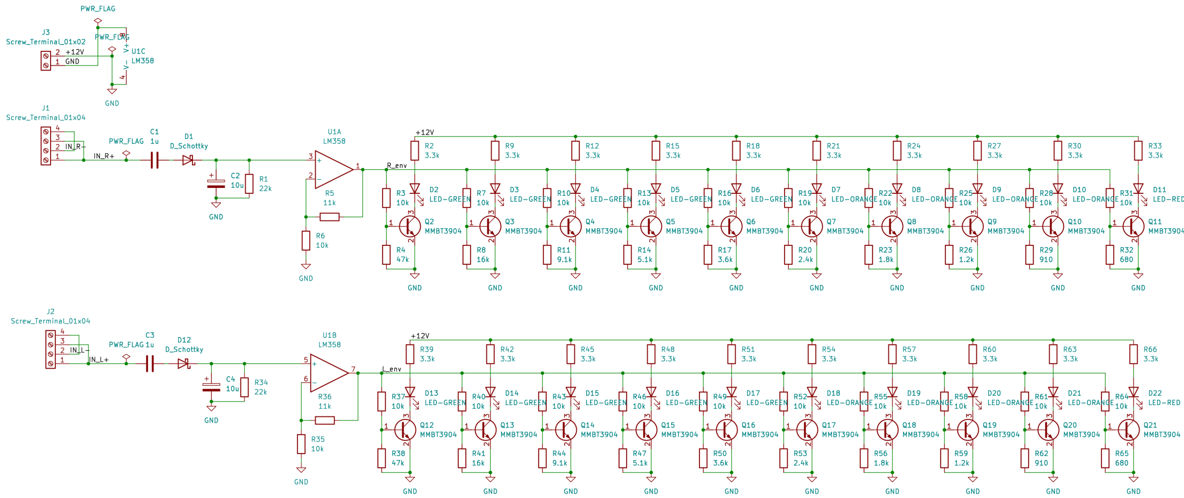

Figure 2: Full schematic: both channels, each half of the LM358 driving its own ten-stage ladder.

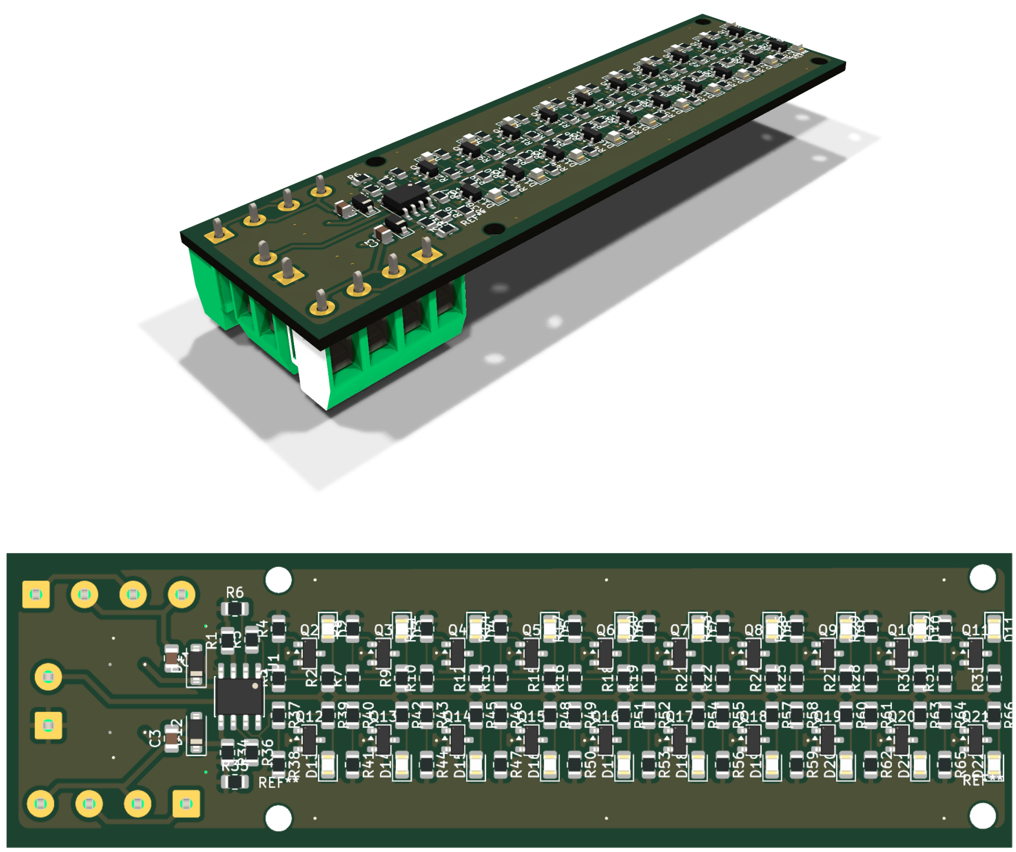

Figure 3: V1 board: two 10-LED channels, MMBT3904 ladder, LM358 front end.